

Same score, different story

How resampling history shapes what spatial-frequency features
detect in AI-generated images

CSE 4883 Digital Image Processing · final presentation

Member 1 · Member 2 · Member 3

Introduction · Novelty · Methodology · Experiments · Results · Limitations · Future

Real and fake images arrive differently

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Real photo



JPEG 500×375
shrunk 0.68×

BigGAN



PNG 128×128
enlarged 2×

ADM



PNG 256×256
unchanged 1×

SD v1.4



PNG 512×512
shrunk 0.5×

File format and size alone separate the classes (AUROC **1.000**). A fair pipeline gives every image 256×256 px, but **each class is resized by a different factor.**

Contribution: a feature-level processing audit

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Established questions

Compression history

JPEG and size bias

Benchmark comparability

content, format, size

Resizing history

resize vs native, JPEG

one level
down

This project · 14 named features

Per feature, per generator

which statistic moves, and for which generator

Two stresses, one fixed protocol

matched JPEG + unseen generator

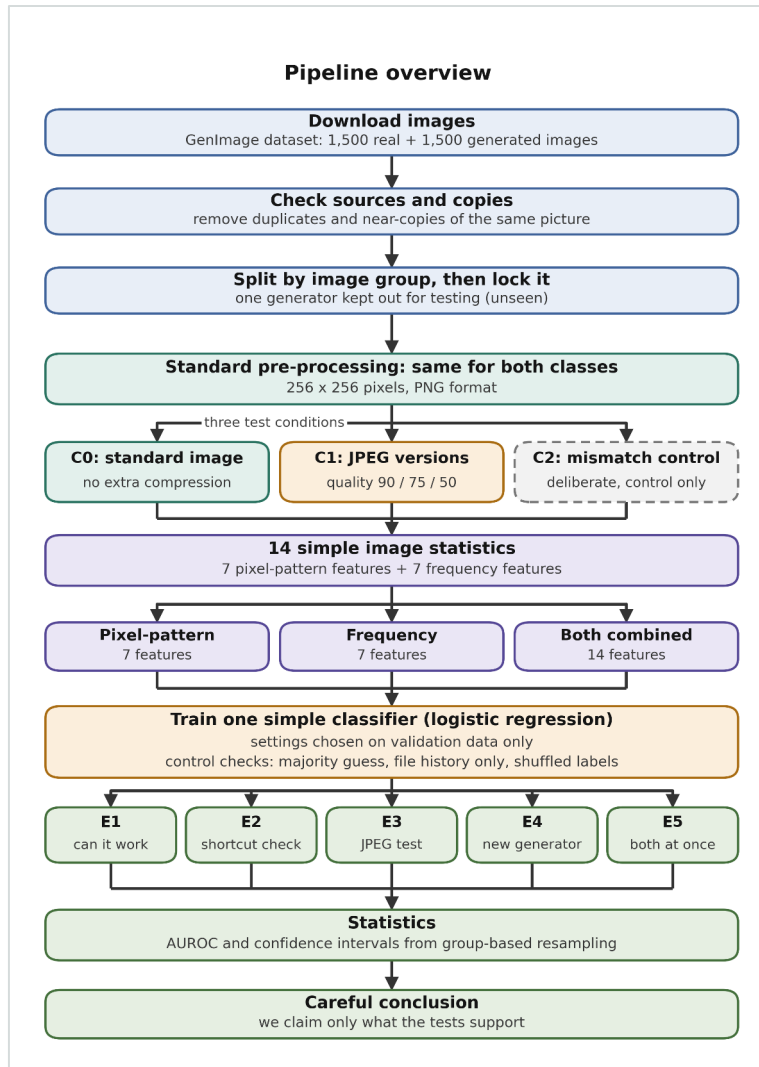
Exploratory reference-data check

BigGAN pixels unchanged; crop and training data change

Contribution: trace processing effects through features, generator scores and decisions

Pipeline, fixed before any image was processed

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1

3,000 GenImage images

1,500 real · 500 each: BigGAN, SD v1.4, ADM

2

Clean before splitting

duplicates, near-copies, odd colour profiles removed

3

Same processing for both classes

resize to 256 × 256, lossless PNG

4

Three conditions

C0 clean · C1 JPEG 90 / 75 / 50 · C2 unfair control

5

Strict statistics

split by image group · one locked 600-image test set · 95% CIs

14 explainable features, one simple classifier

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Pixel arm: 7 features

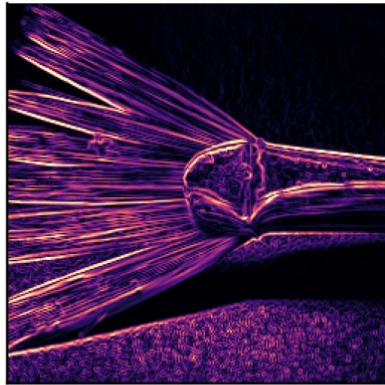
Frequency arm: 7 features

Luminance I



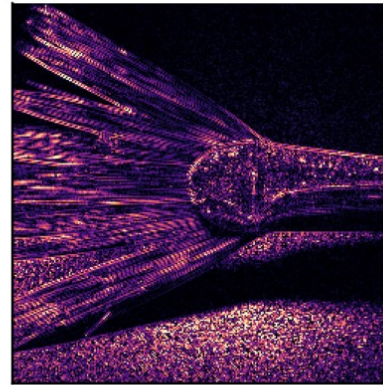
f01-f03
variance, skew, kurtosis

Sobel gradient $|\nabla I|$



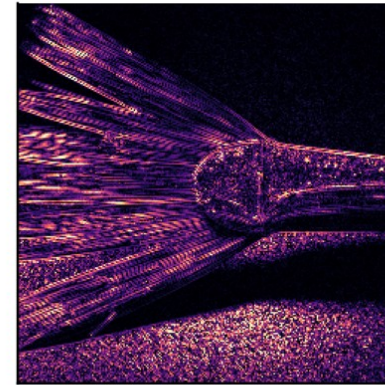
f04-f05
gradient mean, std

Laplacian



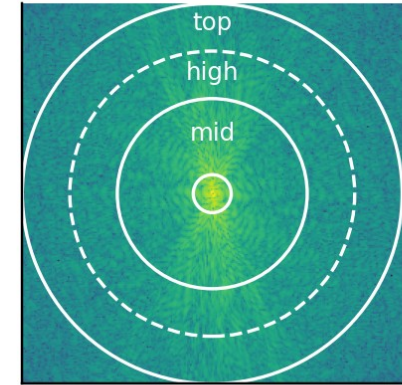
f06
Laplacian variance

High-pass residual



f07
residual variance

Power spectrum (Hann)



f08-f14 band fractions, Nyquist ratio, slope, anisotropy, peak

14 features
per image

Standardise
fit on train only

Logistic regression
L2, C chosen on val

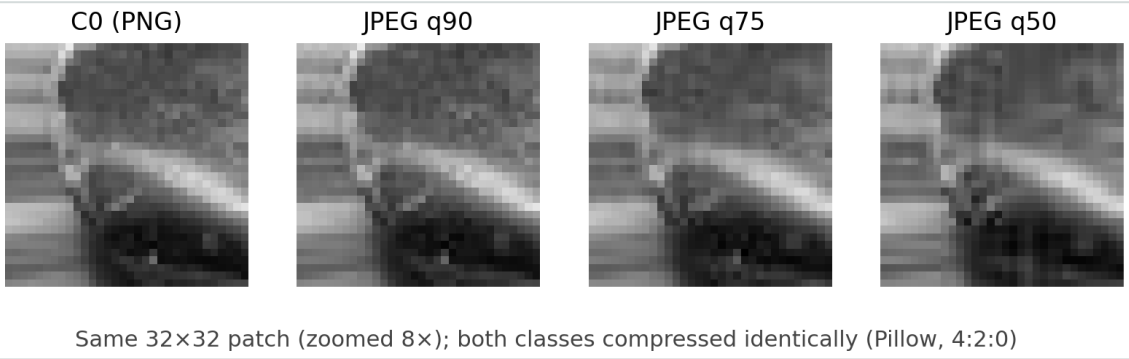
Threshold
best val balanced acc.

Test AUROC
+ 95% bootstrap CI

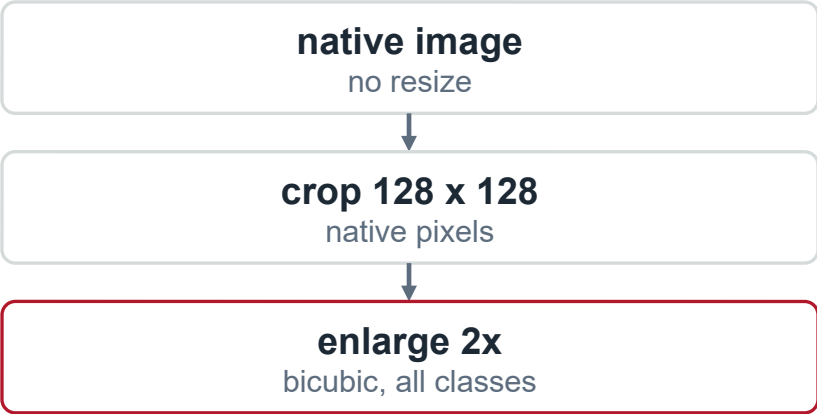
Experiments: what each one tests

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	Images	Trained on	Why it matters
E1	clean (C0)	all 3 generators	is there any signal?
E2	reals JPEG, fakes PNG	all 3	how an unfair pipeline cheats
E3	both JPEG 90 / 75 / 50	all 3	real images are compressed
E4	clean (C0)	SD v1.4 held out	a generator never seen
E5	JPEG 75	SD v1.4 held out	both stresses (headline)
R	native crop + shared 2x resize	as E1 and E4	is it resampling history?



R: exploratory crop + resampling



E1–E5 were fixed in advance.
R was added after the results: exploratory.
BigGAN (128 px) is pixel-identical in C0 and R.

Results: combined wins pooled – not everywhere

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	Pooled	BigGAN	SD v1.4	ADM
E1 pixel	0.666	0.824	0.581	0.591
E1 frequency	0.752	0.997	0.637	0.624
E1 combined	0.793	0.990	0.712	0.678
E5 pixel	0.631	0.888	0.465	0.540
E5 frequency	0.769	0.981	0.568	0.757
E5 combined	0.786	0.969	0.629	0.761

unseen in E5

Test AUROC (0.5 = chance). E5: SD v1.4 never seen in training, JPEG 75.
Source: results/*.csv, results/diagnostics/ (config 87f5489e9d59)

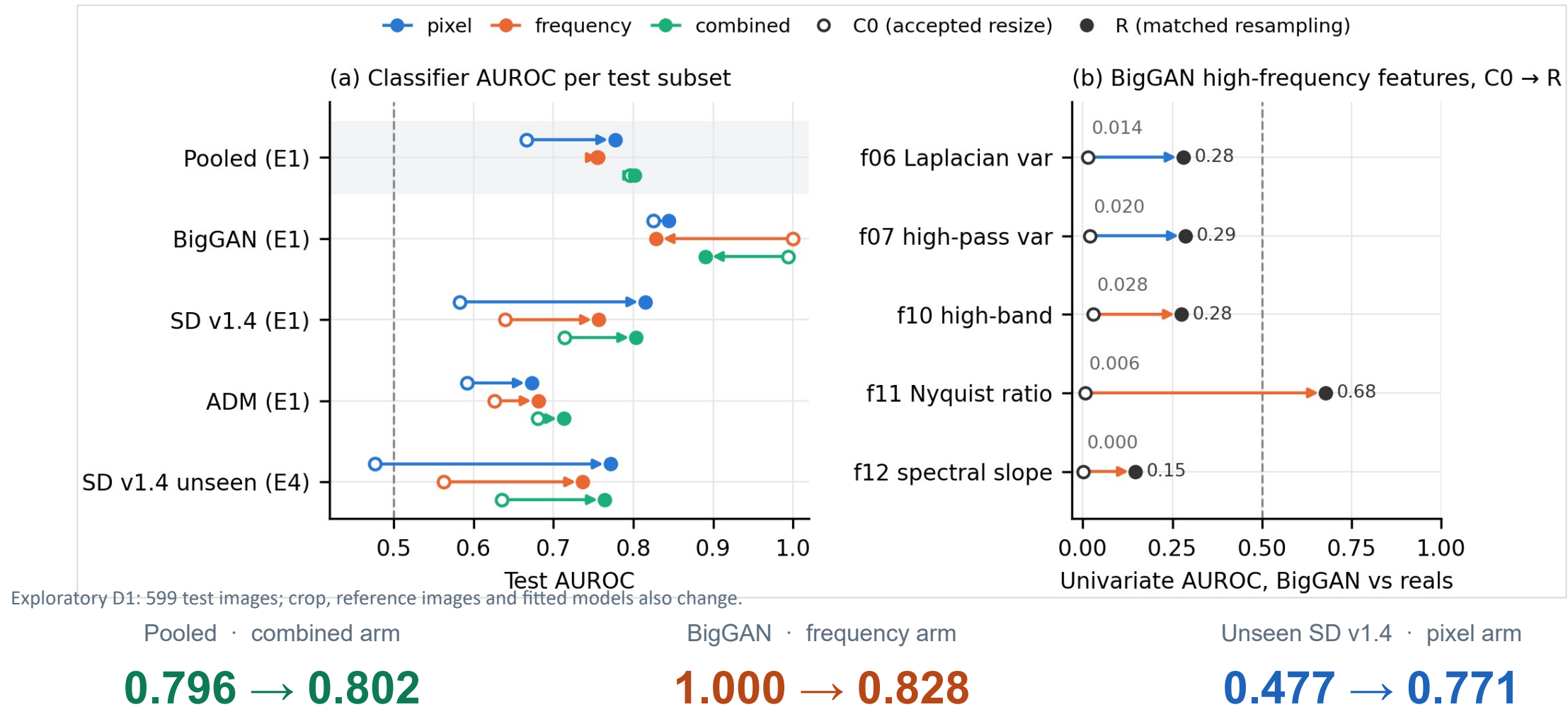
Controls (matched conditions)

Always guess	0.500
File history (matched only)	0.500
Shuffled labels (×200)	0.50–0.52
Original file facts	1.000

Stress	AUROC change
Matched JPEG, any arm	at most 0.025
SD v1.4 unseen, combined	−0.078

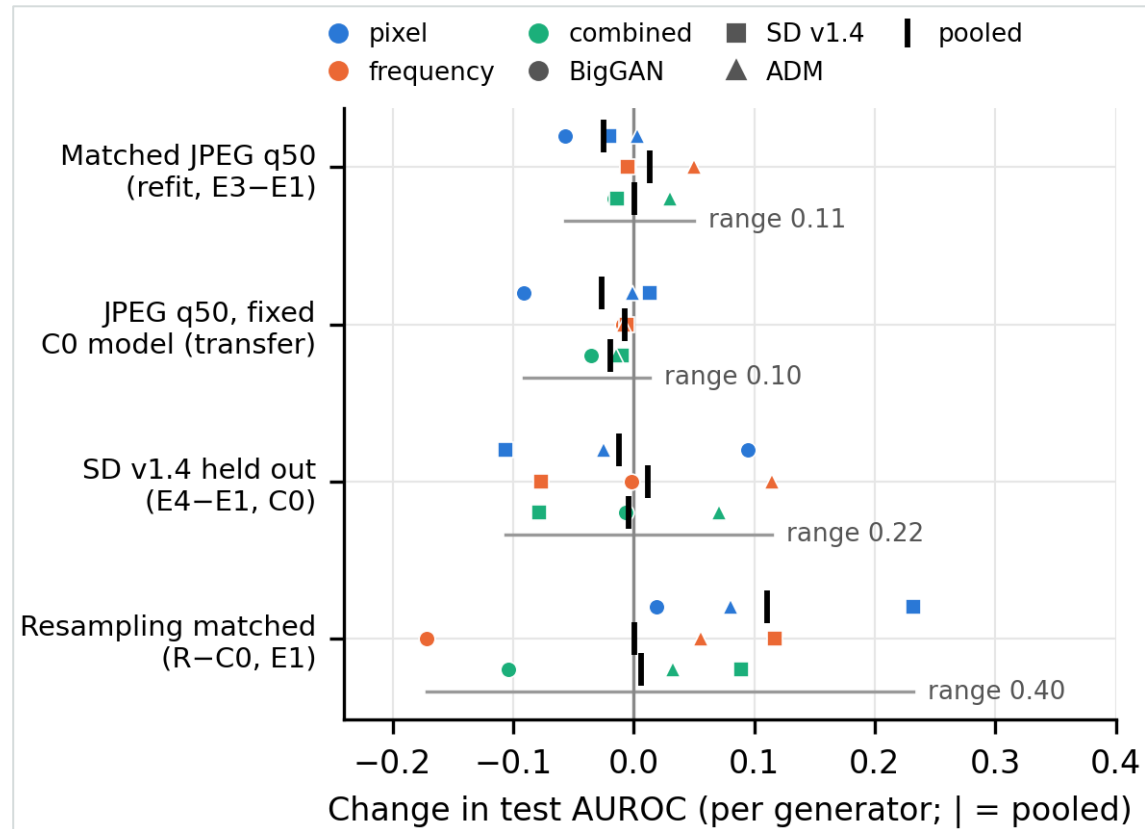
Similar pooled score, different generator results

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Processing changes scores and decisions

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Each dot: one feature set on one generator; tick = pooled.

Which arm looks stronger? pixel – frequency

Pipeline	AUROC difference [95% CI]
Accepted resize	-0.088 [-0.130, -0.045]
Same resampling	+0.022 [-0.015, 0.056]

JPEG q50 on the clean model: reals flagged as fake

Arm	clean	JPEG q50
Pixel	98 / 300	157 / 300
Frequency	14 / 300	16 / 300
Combined	37 / 300	42 / 300

Limitations and planned extensions

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Limitation

R is post-hoc; crop and models also change



Planned extension

pre-specify comparison with matched field of view

only SD v1.4 held out



hold out each generator in turn

real and fake content not matched



content-matched real / fake pairs

one JPEG encoder (Pillow 4:2:0)



a second encoder and subsampling

one dataset, one random split



repeated splits and a second dataset

A pooled score cannot tell you what simple features detect.

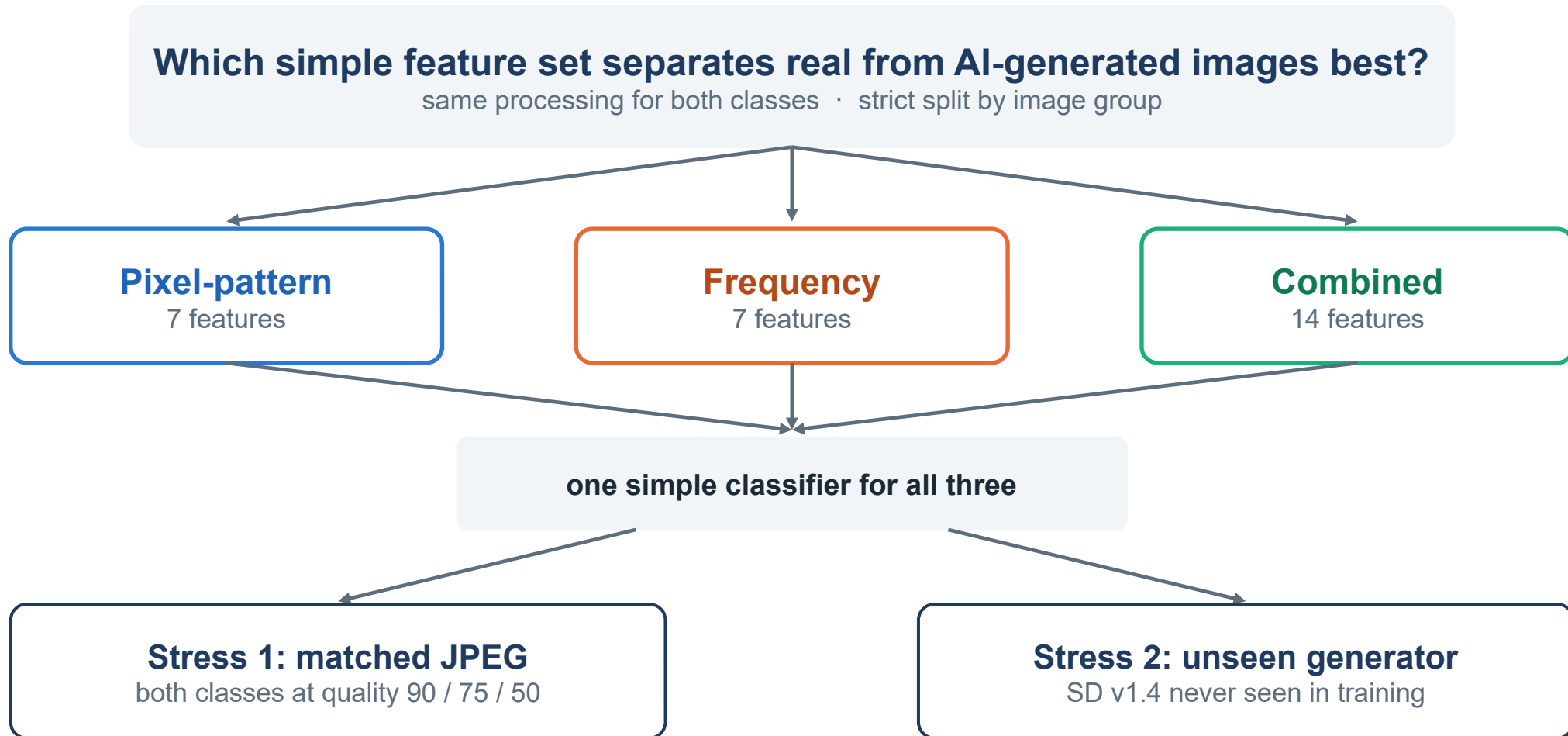
Report resampling history, and report per generator.

Combined features scored best pooled (**0.793**; **0.629** on an unseen generator under JPEG 75), yet the same pooled score hid per-generator changes of **-0.17** to **+0.30**.

Thank you · questions?

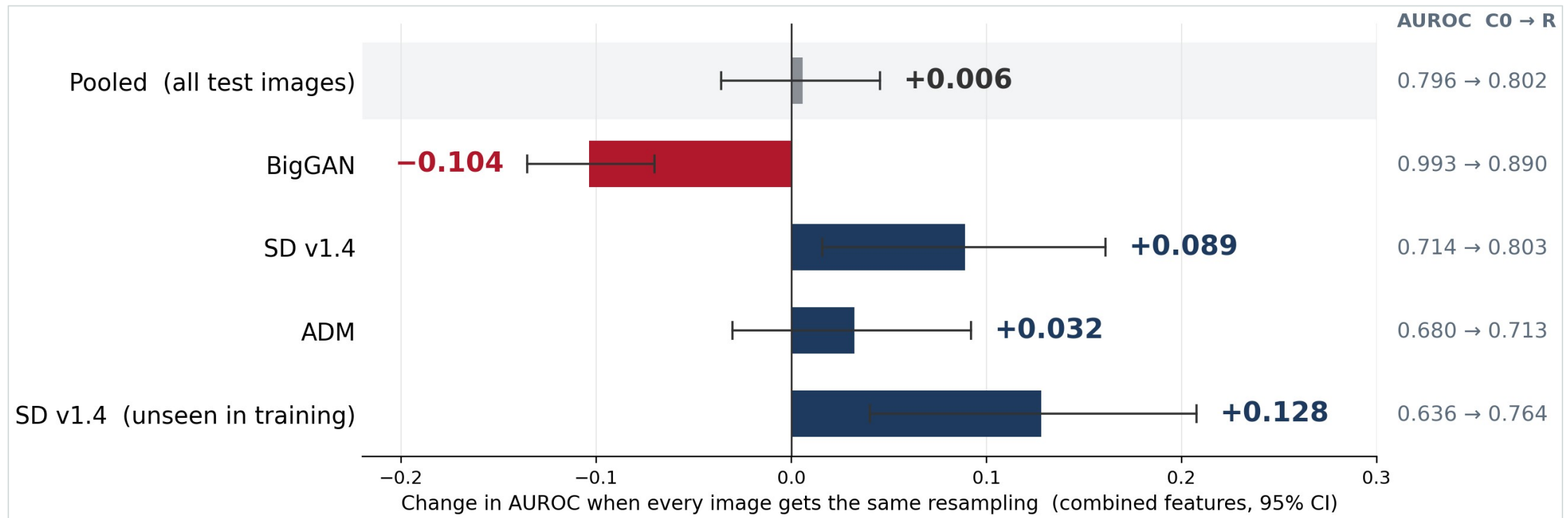
Our research question

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Novelty: same score, different story

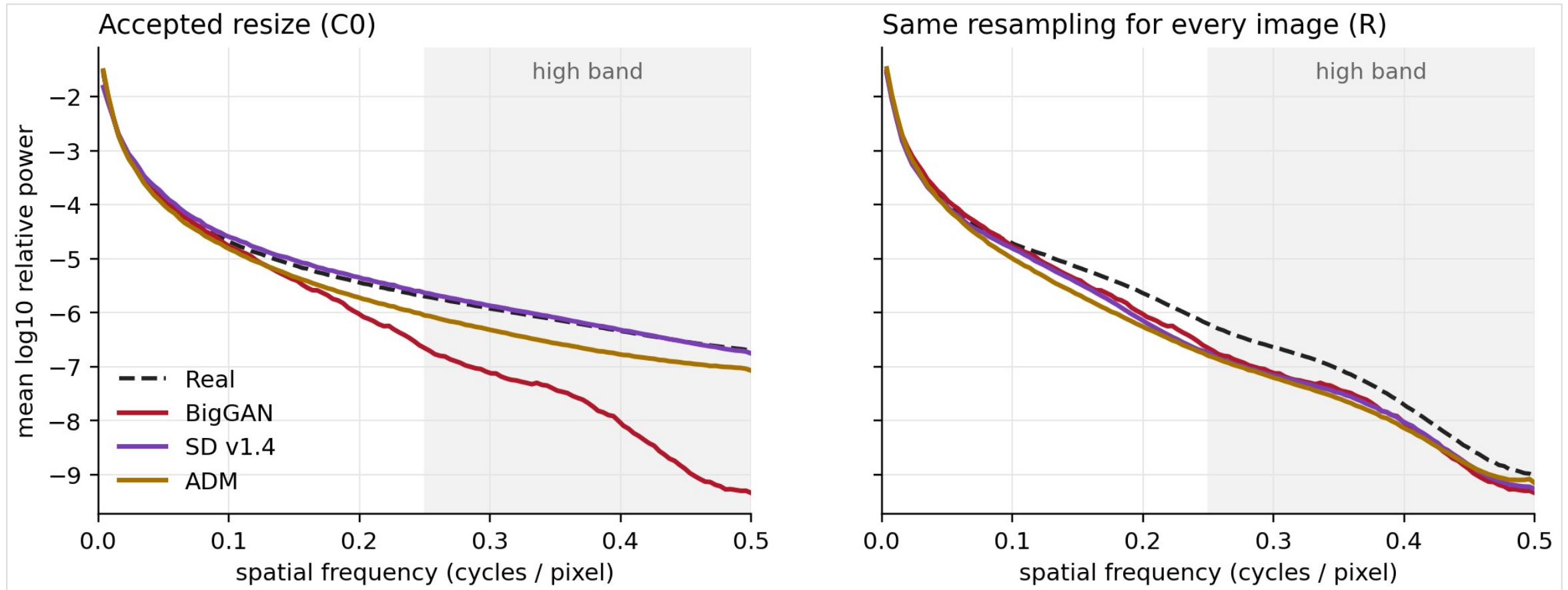
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Pooled score: similar. **Per-generator scores: move both ways.**

BigGAN's high-band deficit is consistent with enlargement

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Mean power spectrum per class, all images. Left: BigGAN, the only class enlarged 2x, loses energy above 0.25 cycles/pixel. Right: with the native crop + shared 2x resize, the three fake curves coincide.

Source: results/diagnostics/radial_spectra.csv